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US Technology | North America

Global Cloud Capex Tracker:
Spending Acceleration
Continues

Cloud capex continues to move higher (+29% Y/Y in CY27, vs. +14% prior) as demand remains ahead of supply. With cloud growth accelerating, AI monetization improving, and hyperscalers expressing growing confidence in ROIC, we believe Cons capex estimates remain too conservative and skewed higher.

Key Takeaways

- All 4 US hyperscalers remain supply constrained, with demand from both external cloud customers and internal AI workloads exceeding capacity.
- Cloud rev growth accelerated despite capacity limits as utilization improves, AI services scale, and AI adoption drives broader infrastructure consumption.
- Hyperscalers expressed greater confidence in CY26-CY27 ROIC, supported by rising backlogs, multi-year commitments, pricing power, and efficiency gains.
- AWS, Azure, and GCP all see sizable runway ahead as enterprise AI adoption remains early and cloud migration continues.
- Strong operating cash flow, equity/debt financing, leasing strategies, ASICs, and infrastructure efficiencies are helping fund capex while easing FCF pressure.

2027 Consensus cloud capex is now tracking to ~\$1.2T, or ~30% Y/Y growth, 15 points, or ~\$170B, higher than pre-C2Q earnings (Exhibit 2). The uptick in 2027 capex is driven by the strong CY26 guidance and CY27 commentary by all 4 US hyperscalers this earnings season (Exhibit 3). We believe Consensus estimate's 29% Y/Y growth in 2027 – representing a 68-point Y/Y deceleration vs. 2026 – is too conservative, as it implies non-AI cloud capex growth of just 7% Y/Y in CY27 (Exhibit 8). We therefore wouldn't be surprised if Consensus cloud capex estimates move higher from here given (1) MSe aggregate cloud cash capex is \$1.4T, 17% above Consensus (Exhibit 7), and (2) aggregate cloud capex estimates for the top 14 spenders has increased every quarter for, at least, the last 10 quarters. In fact, if we go back to the beginning of this year, 2026-2027 combined Global Cloud capex was forecasted at \$1.3T – even lower than the latest MSe for CY27 alone (Exhibit 5). Today, it's forecasted at \$2.1T, or an incremental ~\$800B of spending, equivalent to 33% of aggregate revenue across CY26-CY27, ~4x the 2014-2023 capital spending intensity average (Exhibit 3).

As monthly tokens processed grows exponentially (Exhibit 7), aggregate cloud revenue for GCP/AWS/Azure accelerates (Exhibit 8), data center commitments

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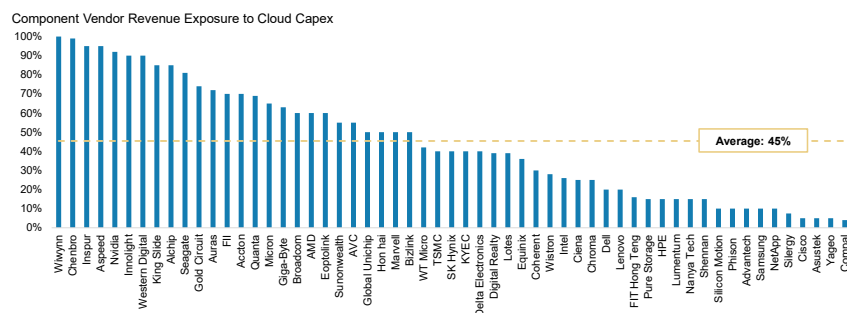
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expand, and data center component suppliers highlight accelerating demand and elongating visibility, we believe there will continue to be upward pressure on hyperscaler capex estimates. For a full list of MS-covered data center component vendors with exposure to Global Cloud Capex that we believe will benefit from this trend, please see [Exhibit 1](#).

Exhibit 1: Below, we detail the estimated revenue exposure to cloud capex for 50+ different technology companies across Morgan Stanley's US, European and Asian equity research coverage



Source: Company data, Morgan Stanley Research estimates

5 key takeaways from US hyperscaler C2Q26 earnings:

1. 3 of the 4 US hyperscalers raised their CY26 capex spending. **GOOGL** lifted its CY26 capex guide to \$195-\$205B (from \$180-\$190B prior), citing an acceleration in capacity delivery to meet demand that continues to outpace supply. **AMZN** made the most dramatic move, bumping its CY26 cash capex to ~\$220B from ~\$200B – with memory cost inflation as the culprit – and still flagged it won't be enough to satisfy demand. **META** narrowed its range to \$130-\$145B (from \$125-\$145B), staying demand-constrained and focused on maximizing '26/'27 capacity. **MSFT**'s underlying CY26 spend is unchanged at ~\$190B, though a lease accounting reclassification (shifting from finance to operating leases) mechanically lowers the reported figure to ~\$175B.

2. Compute demand remains stronger than available supply. All 4 US hyperscalers highlighted ongoing capacity constraints, with **GOOGL** noting demand continues to exceed internal supply, **MSFT** reiterating that Azure demand remains above available capacity, **AMZN** stating 2026 capacity will remain insufficient to meet demand and that much of its 2027 capacity is already reserved, and **META** expecting industry-wide compute availability to remain tight for the foreseeable future. Importantly, demand is coming from both external cloud customers and internal AI workloads, suggesting limited signs of moderation despite the unprecedented pace of infrastructure deployment.

3. Cloud revenue growth is accelerating despite supply constraints. Cloud revenue growth is no longer solely a function of incremental capacity additions, but also improving utilization, higher-value AI services, and expanding AI-driven consumption of broader cloud infrastructure. For context, **Azure** accelerated to +43% Y/Y growth with management attributing upside to efficiency gains across CPU/GPU fleets, process improvements, and faster deployment of new capacity, while Azure's incremental capacity was monetized almost immediately. **GCP**

accelerated to +82% Y/Y growth, driven by enterprise AI products, AI infrastructure services, core GCP workloads, and growing contributions from TPU system sales.

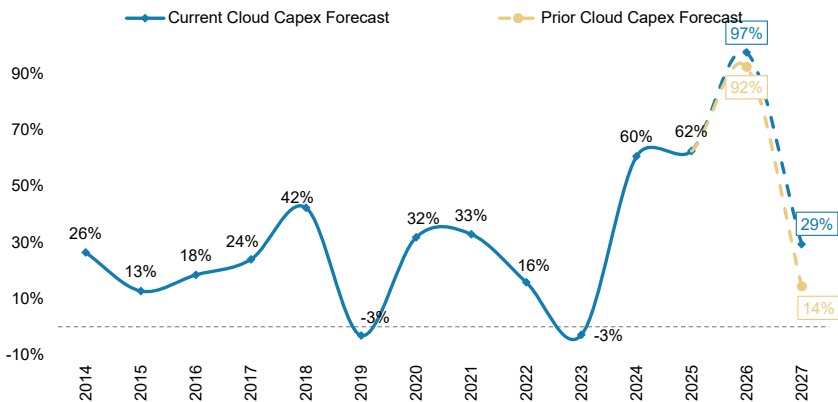
AWS delivered its fastest growth rate (+37% Y/Y) in 18 quarters, supported by both AI and traditional cloud workloads, with mgmt emphasizing that AI adoption is increasingly driving incremental consumption across storage, databases, networking, and core cloud services.

4. Confidence in CY26-CY27 ROIC continues to improve. Across hyperscalers, mgmt commentary suggests conviction around future returns remains high despite rapidly rising investment levels. **GOOGL** highlighted strong long-term demand indicators, growing backlog commitments, renewed contracts and attractive returns from serving customers even when using higher-cost 3rd-party capacity. **MSFT** pointed to improving confidence driven by TAM expansion, infrastructure efficiencies, application-layer optimization, silicon innovation, model diversification, and pricing power. **AMZN** emphasized that most AI capacity is contracted on multi-year terms, with server investments achieving breakeven in <3 years and meaningful FCF generation thereafter, while mgmt now sees a path toward a substantially larger long-term AWS revenue opportunity. **META** similarly highlighted confidence in monetizing AI infrastructure across consumer experiences, enterprise offerings, APIs, agentic solutions, and direct compute monetization, while maintaining flexibility to optimize future deployment decisions.

5. Funding sources continue to diversify, while FCF pressure is being actively addressed. Hyperscalers continue to rely primarily on operating cash flow to fund incremental AI investments, but are supplementing internal funding with additional balance-sheet flexibility. **GOOGL** highlighted cash flow, debt financing, and recent equity issuance as sources of capital, while **META** pointed to strong cash generation, incremental debt financing, and infrastructure partnerships. **AMZN** noted debt issuance and additional financing flexibility as needed. Meanwhile, mgmt teams are employing several strategies to mitigate pressure on FCF, including extending asset useful lives (**MSFT**), increasing the use of operating leases (**MSFT**), utilizing 3rd-party capacity as a bridge while internal capacity ramps (**GOOGL**), leaning further into custom silicon and infrastructure efficiencies (**GOOGL** and **AMZN**), and maintaining a high proportion of shorter-duration server investments that can be adjusted if demand changes (**MSFT**). Taken together, these comments suggest that hyperscalers remain focused not only on maximizing compute availability, but also on preserving balance sheet flexibility while navigating an extended investment cycle.

Exhibit 2: Post US hyperscaler C2Q26 earnings, our cloud capex tracker is now pointing to 29% Y/Y in 2027, up 15 points from the prior forecast of +14% Y/Y about a month ago

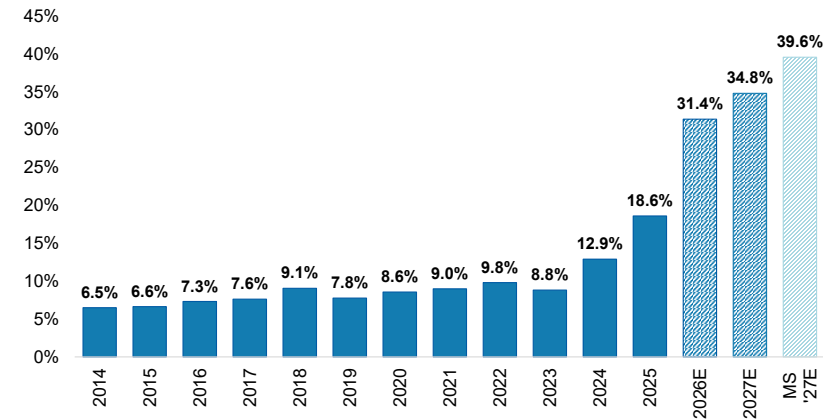
Top 14 Cloud Providers: Cloud Capex Y/Y Growth



Source: Company data, FactSet, Morgan Stanley Research. Note: Cloud capex includes capex from Alphabet, Amazon, Microsoft, Meta Platforms, Alibaba, Tencent, Baidu, CoreWeave, Nebius, IREN, Tesla, Apple, IBM, and Oracle. We start to include NVIS from 2023, and CRWW/IREN from 2024.

Exhibit 3: With stronger cloud capex now expected in 2027, overall capital intensity is now approaching 35% based on Consensus estimates, a new all-time high

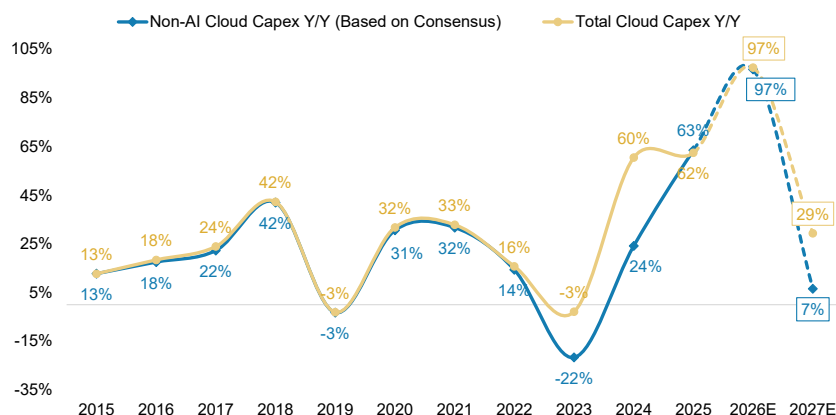
Top 14 Cloud Providers: Capital Intensity



Source: Company data, Morgan Stanley Research estimates

Exhibit 4: Based on Consensus, non-AI cloud capex growth are accelerating ~90 points to just +7% Y/Y in CY27, which we view as too conservative.

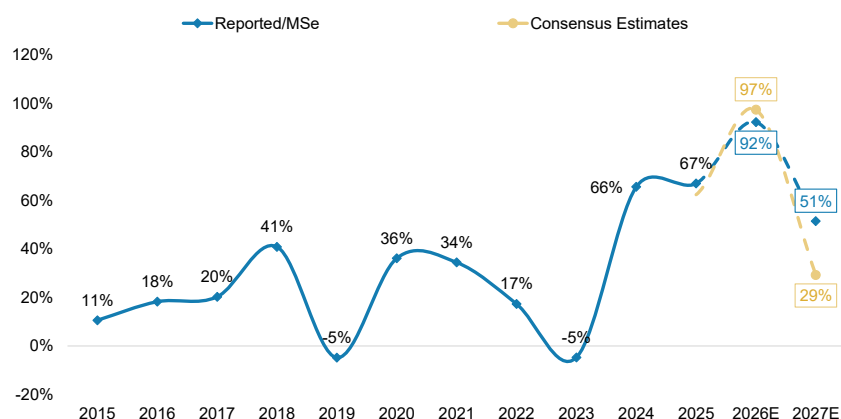
Top 14 Cloud Providers: Non-AI Cloud Capex Y/Y Growth



Source: Company data, FactSet, Morgan Stanley Research estimates. Note: Non-AI cloud capex = cloud capex - AI GPU/ASIC revenues from NVDA/AMD/INTC/AVGO/MRVL.

Exhibit 5: MSe is pointing to more than 50% Y/Y of cloud capex growth in CY27, 22 points higher than Consensus.

Top 14 Cloud Providers: Cash Capex Y/Y Growth



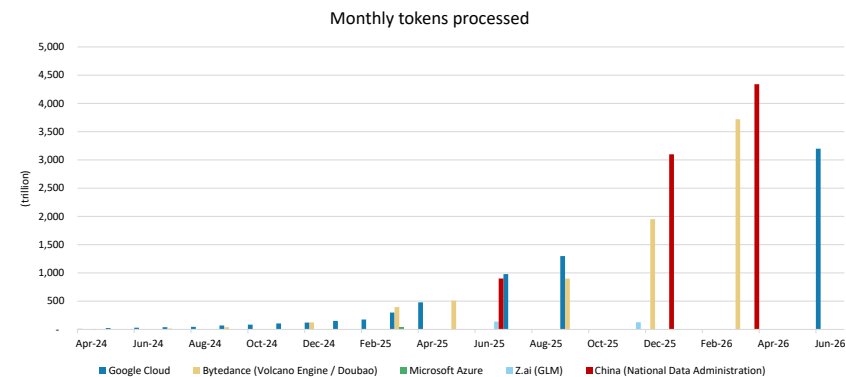
Source: Company data, FactSet, Morgan Stanley Research estimates.

Exhibit 6: Consensus CY27 capex just increased by 70%+ (or ~\$500B) YTD, which makes us believe that as long as the qualitative commentary on cloud spending stays robust, there likely will be further upward revisions to CY27 capex

Company	Current CY27 Capex	CY27 Capex Pre-Q4 Earnings	CY27 Capex Beginning of 2026	CY26 Capex Beginning of 2026	% Change vs. Pre-Q4 Earnings	\$ Change vs. Pre-Q4 Earnings	% Change vs. Beginning of 2026	\$ Change vs. Beginning of 2026
Google	\$297.3	\$256.9	\$129.4	\$114.3	15.7%	\$40.4	129.8%	\$167.9
Amazon	\$253.7	\$231.5	\$160.5	\$145.4	9.6%	\$22.2	58.1%	\$93.3
Microsoft	\$204.4	\$198.2	\$120.1	\$107.4	3.1%	\$6.2	70.2%	\$84.3
Meta Platforms	\$206.9	\$171.5	\$122.0	\$109.0	20.6%	\$35.4	69.6%	\$84.9
Alibaba	\$21.5	\$21.4	\$18.5	\$17.8	0.4%	\$0.1	16.4%	\$3.0
Tencent	\$21.6	\$21.5	\$15.6	\$14.1	0.4%	\$0.1	39.1%	\$6.1
Baidu	\$2.4	\$2.3	\$1.6	\$1.6	2.2%	\$0.1	46.2%	\$0.7
CoreWeave*	\$39.9	\$39.3	\$22.3	\$26.7	1.6%	\$0.6	79.0%	\$17.6
Nebius	\$26.4	\$26.5	\$15.4	\$11.5	(0.4%)	(\$0.1)	71.7%	\$11.0
IREN	\$6.3	\$6.5	\$5.3	\$5.1	(4.5%)	(\$0.3)	18.7%	\$1.0
Tesla	\$22.7	\$21.2	\$12.2	\$11.1	7.2%	\$1.5	85.5%	\$10.5
Apple	\$14.2	\$13.7	\$16.3	\$14.4	3.2%	\$0.4	(7.2%)	-\$1.1
IBM	\$1.5	\$1.6	\$1.6	\$1.6	(6.6%)	(\$0.1)	(7.0%)	-\$0.1
Oracle	\$97.4	\$92.2	\$64.3	\$53.2	5.7%	\$5.3	51.4%	\$33.1
Total	\$1,216.1	\$1,194.3	\$703.9	\$633.2	10.1%	\$111.8	72.8%	\$512.2
Total ex Amazon	\$962.4	\$872.8	\$543.5	\$487.8	10.3%	\$89.5	77.1%	\$418.9

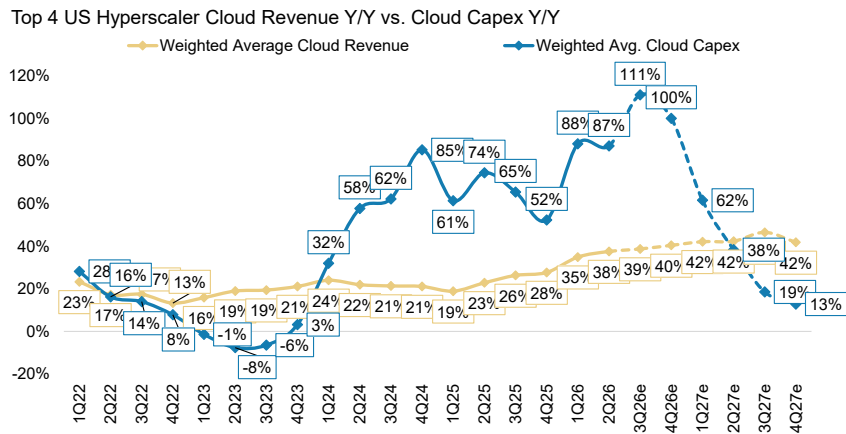
Source: Company data, FactSet, Morgan Stanley Research. Note: Using 5/13/2025 CRWW estimates for 'beginning of 2025' and 'a year ago' as CRWW went public in March '25.

Exhibit 7: Monthly tokens processed by major CSPs suggest that AI inference demand continues to grow exponentially.



Source: Company information, Morgan Stanley Research. Note: Bytedance numbers represent monthly token use run-rate based on end-of-month daily tokens

Exhibit 8: US top 4 hyperscaler cloud revenue growth is expected to accelerate to the 40-45% range over the next several quarters, strongest since 2020.



Source: Company data, Morgan Stanley Research estimates

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(as of July 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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INDUSTRY COVERAGE: IT Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Erik W Woodring		
Apple, Inc. (AAPL.O)	O (05/26/2009)	\$308.91
CDW Corporation (CDW.O)	O (06/23/2026)	\$147.81
Cricut Inc (CRCT.O)	U (08/12/2021)	\$4.69
Dell Technologies Inc. (DELL.N)	E (06/01/2026)	\$405.37
Everpure, Inc. (P.N)	E (06/11/2024)	\$77.17
Garmin Ltd (GRMN.N)	E (02/18/2026)	\$293.78
GoPro Inc (GPRO.O)	U (12/12/2023)	\$0.75
Hewlett Packard Enterprise (HPE.N)	E (11/16/2025)	\$47.90
HP Inc. (HPQ.N)	U (11/16/2025)	\$27.27
IBM (IBM.N)	E (01/18/2023)	\$223.65
Ingram Micro (INGM.N)	E (06/11/2025)	\$28.21
Kornit Digital Ltd. (KRNT.O)	E (11/06/2025)	\$15.22
Logitech International SA (LOGI.O)	U (01/20/2026)	\$102.48
NetApp Inc (NTAP.O)	U (01/20/2026)	\$178.50
Resideo Technologies Inc (REZI.N)	O (08/11/2025)	\$34.30
Seagate Technology (STX.O)	O (03/26/2024)	\$856.13
SmartRent, Inc. (SMRT.N)	++	\$1.01
Sonos Inc. (SONO.O)	E (11/06/2025)	\$14.66
TD Synnex Corporation (SNX.N)	O (06/11/2025)	\$255.70
Teradata (TDC.N)	O (04/08/2025)	\$30.99
Western Digital (WDC.O)	O (04/16/2025)	\$544.84

Sanjit K Singh

Nutanix Inc (NTNX.O)

E (01/12/2026)

\$59.01

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Software

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Adam Wood		
Adobe Inc. (ADBE.O)	U (07/21/2026)	\$250.41
Atlassian Corporation PLC (TEAM.O)	O (01/13/2020)	\$101.02
Intuit (INTU.O)	E (07/21/2026)	\$316.07
Microsoft (MSFT.O)	O (07/21/2026)	\$464.72
Salesforce, Inc. (CRM.N)	E (07/21/2026)	\$184.02
Samsara Inc (IOT.N)	E (03/23/2023)	\$37.27
ServiceNow Inc (NOW.N)	O (09/24/2025)	\$111.23
Shopify Inc (SHOP.O)	O (04/19/2024)	\$117.15
Workday Inc (WDAY.O)	U (07/21/2026)	\$160.34
Chris Quintero		
BILL Holdings Inc (BILL.N)	E (06/10/2025)	\$45.13
Blackline Inc (BL.O)	E (07/21/2026)	\$31.40
Box Inc (BOX.N)	E (05/21/2024)	\$31.55
CCC Intelligent Solutions Holdings Inc (CCC.O)	O (11/13/2024)	\$6.09
Descartes Systems Group Inc (DSGX.O)	O (01/15/2026)	\$75.16
DocuSign Inc (DOCU.O)	E (01/16/2024)	\$54.83
Manhattan Associates Inc. (MANH.O)	E (10/21/2025)	\$191.36
Navan Inc (NAVN.O)	O (11/24/2025)	\$26.62
ServiceTitan Inc (TTAN.O)	O (01/20/2026)	\$83.06
SPS Commerce Inc (SPSC.O)	U (07/21/2026)	\$73.39
Vertex Inc. (VERX.O)	E (07/21/2026)	\$12.91
Via Transportation Inc (VIA.N)	O (01/20/2026)	\$20.34
Elizabeth Porter, CFA		
Amplitude Inc. (AMPL.O)	O (01/15/2026)	\$8.88
Asana Inc (ASAN.N)	U (05/20/2025)	\$8.32
Autodesk (ADSK.O)	O (08/23/2024)	\$234.20
Figma Inc (FIG.N)	E (08/25/2025)	\$24.32
Five9 Inc (FIVN.O)	E (10/10/2022)	\$27.51
Freshworks Inc (FRSH.O)	E (10/18/2021)	\$11.36
GoDaddy Inc (GDDY.N)	E (07/19/2021)	\$82.74
HubSpot, Inc. (HUBS.N)	O (03/21/2023)	\$237.36
Klaviyo, Inc (KVYO.N)	O (04/29/2026)	\$17.84
Liveramp Holdings Inc (RAMP.N)	E (01/13/2025)	\$37.87
monday.com Ltd (MNDY.O)	O (08/12/2025)	\$87.15
NICE Ltd. (NICE.O)	E (07/21/2026)	\$98.79
RingCentral Inc (RNG.N)	E (08/08/2023)	\$55.62
Sprinklr Inc (CXM.N)	E (07/19/2021)	\$6.36
Sprout Social Inc (SPT.O)	E (11/17/2020)	\$8.58
Toast, Inc. (TOST.N)	O (12/16/2021)	\$32.27
Twilio Inc (TWLO.N)	O (02/24/2025)	\$197.35
Wix.Com Ltd (WIX.O)	E (07/21/2026)	\$55.03
Zeta Global Holdings Corp (ZETA.N)	E (08/01/2024)	\$21.60
Zoom Communications (ZM.O)	E (10/11/2022)	\$96.07
ZoomInfo Technologies Inc (GTM.O)	E (02/01/2024)	\$3.30
Meryl R Thomas, CFA		
Lightspeed Commerce Inc. (LSPD.N)	E (02/18/2021)	\$10.11
Meta A Marshall		

Check Point Software Technologies Ltd. (CHKP.O)	E (10/16/2023)	\$127.13
CrowdStrike Holdings Inc (CRWD.O)	O (03/10/2026)	\$190.86
Fortinet Inc. (FTNT.O)	E (07/21/2026)	\$161.95
Gen Digital Inc. (GEN.O)	E (06/07/2024)	\$27.45
Netskope, Inc. (NTSK.O)	O (10/13/2025)	\$12.59
Okta, Inc. (OKTA.O)	O (12/02/2024)	\$141.93
Palo Alto Networks Inc (PANW.O)	O (10/10/2017)	\$331.83
Qualys Inc (QLYS.O)	U (02/09/2021)	\$144.75
Rapid7 Inc (RPD.O)	U (07/21/2026)	\$9.65
SailPoint Inc (SAIL.O)	O (09/02/2025)	\$16.77
SentinelOne, Inc. (S.N)	E (12/02/2024)	\$19.06
Tenable Holdings Inc (TENB.O)	E (12/02/2024)	\$32.64
Varonis Systems, Inc. (VRNS.O)	E (01/26/2026)	\$39.95
Zscaler Inc (ZS.O)	E (04/22/2026)	\$151.20
Ryan Lountzis		
CoreWeave (CRWV.O)	E (04/22/2025)	\$71.77
Nebius Group NV (NBIS.O)	E (01/15/2026)	\$190.41
Sanjit K Singh		
Akamai Technologies, Inc. (AKAM.O)	O (01/12/2026)	\$115.18
Appian Corp (APPN.O)	E (04/30/2026)	\$26.96
C3.ai (AI.N)	U (01/04/2021)	\$9.18
Cloudflare Inc (NET.N)	O (12/02/2024)	\$278.98
Datadog, Inc. (DDOG.O)	O (01/12/2026)	\$267.97
DigitalOcean Holdings Inc (DOCN.N)	O (01/16/2025)	\$117.50
Dynatrace Inc (DT.N)	E (02/13/2024)	\$44.32
Elastic NV (ESTC.N)	E (07/21/2026)	\$65.79
GitLab Inc (GTLB.O)	E (01/12/2026)	\$34.51
JFrog Ltd. (FROG.O)	E (07/21/2026)	\$79.79
MongoDB Inc (MDB.O)	O (04/12/2023)	\$337.48
Oracle Corporation (ORCL.N)	E (01/15/2019)	\$129.87
PagerDuty, Inc. (PD.N)	U (07/21/2026)	\$10.58
Palantir Technologies Inc. (PLTR.O)	E (02/04/2025)	\$123.06
Snowflake Inc. (SNOW.N)	O (06/24/2025)	\$293.28
UiPath Inc (PATH.N)	E (09/07/2022)	\$12.76

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Internet

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Brian Nowak, CFA		
Airbnb Inc (ABNB.O)	U (12/06/2022)	\$151.52
Alphabet Inc. (GOOGL.O)	O (08/11/2015)	\$356.13
Amazon.com Inc (AMZN.O)	O (04/24/2015)	\$271.58
Booking Holdings Inc (BKNG.O)	O (02/23/2026)	\$192.90
DoorDash Inc (DASH.O)	O (02/21/2024)	\$196.16
Expedia Inc. (EXPE.O)	E (01/09/2019)	\$294.74
Instacart (CART.O)	E (01/29/2024)	\$44.60
Lyft Inc (LYFT.O)	E (10/24/2019)	\$15.86
Meta Platforms Inc (META.O)	O (03/20/2023)	\$556.71
Pinterest Inc (PINS.N)	O (07/20/2025)	\$24.01
Reddit Inc (RDDT.N)	O (12/08/2024)	\$140.67
Snap Inc. (SNAP.N)	E (07/22/2024)	\$4.69
Uber Technologies Inc (UBER.N)	++	\$70.36

Matthew Cost

AppLovin Corp (APP.O)	O (04/10/2025)	\$395.90
Compass, Inc. (COMP.N)	E (01/12/2026)	\$11.39
Criteo SA (CRO.O)	E (01/26/2016)	\$21.82
DoubleVerify Holdings Inc (DV.N)	E (06/25/2024)	\$11.25
Electronic Arts Inc (EA.O)	E (08/04/2021)	\$209.86
Liftoff Mobile Inc. (LFTO.O)	E (06/29/2026)	\$22.20
MNTN Inc (MNTN.N)	E (06/16/2025)	\$9.22
Opendoor Technologies Inc (OPEN.O)	E (07/24/2023)	\$3.77
Playtika Holding Corp (PLTK.O)	E (11/27/2022)	\$3.99
Roblox Corporation (RBLX.N)	O (11/04/2024)	\$35.60
Take-Two Interactive Software (TTWO.O)	O (02/01/2018)	\$242.92
Trade Desk Inc (TTD.O)	E (09/10/2025)	\$18.04
Unity Software Inc (U.N)	O (09/02/2024)	\$31.71
Webtoon Entertainment Inc (WBTN.O)	E (07/22/2024)	\$9.12
Yelp Inc (YELP.N)	U (01/10/2019)	\$26.43
Zillow Group Inc (Z.O)	E (04/18/2018)	\$34.06
Nathan Feather		
Bumble Inc. (BMBL.O)	E (03/08/2021)	\$2.85
Chewy Inc (CHWY.N)	O (10/31/2023)	\$22.60
Duolingo Inc (DUOL.O)	E (02/27/2026)	\$134.81
eBay Inc (EBAY.O)	O (04/18/2024)	\$114.01
Etsy Inc (ETSY.N)	E (07/20/2025)	\$81.68
FIGS, Inc. (FIGS.N)	E (02/29/2024)	\$10.70
Grindr Inc. (GRND.N)	O (07/01/2026)	\$17.34
Match Group Inc (MTCH.O)	E (04/18/2024)	\$39.41
Peloton Interactive, Inc. (PTON.O)	E (03/14/2022)	\$6.39
Revolve Group Inc (RVLV.N)	E (10/20/2024)	\$25.11
WW International Inc (WW.O)	E (08/01/2025)	\$14.36

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/03/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK\$40.28
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb156.65
BYD Electronics (0285.HK)	O (04/28/2023)	HK\$24.24
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb8.15
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb16.80
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb394.08
Genius Electronic Optical Co. Ltd. (3406.TW)	E (04/23/2025)	NT\$488.00
GoerTek Inc (002241.SZ)	U (04/23/2025)	Rmb23.24
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.70
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb37.95
Largan Precision (3008.TW)	O (07/10/2026)	NT\$3,960.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb6.88
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb12.46
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb53.71
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb7.35
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK\$7.02
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb45.89
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK\$33.90
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb63.00
Sunny Optical (2382.HK)	E (03/23/2026)	HK\$56.55

Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb180.50
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.42
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK\$27.96
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb262.30
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK\$95.70
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb89.21
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.71
Zhongji Innolight Co Ltd (300308.SZ)	++	Rmb902.50
ZTE Corporation (0763.HK)	O (06/04/2026)	HK\$25.26
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb34.02
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT\$2,335.00
Advantech (2395.TW)	O (01/20/2021)	NT\$576.00
AirTAC International (1590.TW)	O (04/16/2025)	NT\$1,415.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT\$2,550.00
AU Optronics (2409.TW)	E (02/10/2026)	NT\$23.90
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT\$944.00
Bizlink (3665.TW)	O (03/10/2025)	NT\$2,095.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.42
Chenbro (8210.TW)	O (07/23/2025)	NT\$1,095.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT\$1,960.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT\$1,580.00
E Ink Holdings Inc. (8069.TWO)	O (05/11/2026)	NT\$189.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT\$50.40
Fositek Corp (6805.TW)	O (06/25/2025)	NT\$1,470.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT\$332.00
Innolux (3481.TW)	E (04/07/2025)	NT\$45.80
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT\$8,635.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb31.57
Lite-On Technology (2301.TW)	E (01/15/2025)	NT\$229.50
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.70
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.53
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb167.00
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT\$28.80
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT\$799.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT\$182.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT\$36.35
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK\$4.90
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb55.90
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT\$324.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT\$872.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT\$253.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb71.66
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT\$1,790.00
Lenovo (0992.HK)	O (07/10/2026)	HK\$24.74
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT\$2,005.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT\$953.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT\$86.60
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT\$296.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb102.03
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb302.31
Unimicron (3037.TW)	O (02/23/2026)	NT\$865.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT\$321.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT\$186.00

Wiwynn Corp (6669.TW)	O (11/10/2025)	NT\$5,925.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT\$552.00
Zhen Ding (4958.TW)	O (05/18/2026)	NT\$421.50

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/03/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	\$78.47
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb309.85
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$115.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,250.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$610.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,028.00
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$4,185.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$866.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$419.50
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$126.80
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$428.20
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$233.50
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb67.02
MediaTek (2454.TW)	O (11/28/2025)	NT\$3,910.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb665.00
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$396.50
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb641.30
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb88.22
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$1,760.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb94.40
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$424.50
SMIC (0981.HK)	O (10/21/2025)	HK\$62.20
TSMC (2330.TW)	O (02/07/2022)	NT\$2,370.00
UMC (2303.TW)	O (05/19/2026)	NT\$118.00
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$151.00
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$311.00
Daisy Dai, CFA		
ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$150.30
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb51.84
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb87.05
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb27.45
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb261.00
Innoscence (2577.HK)	E (10/13/2025)	HK\$40.58
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb61.00
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$24.16
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb81.39
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb80.13
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb64.10
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb22.02
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb79.98
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$733.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,340.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$15,975.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$94.70

Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$156.50
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb103.70
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb340.74
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$110.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$248.80
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb189.94
Novatek (3034.TW)	U (02/04/2026)	NT\$519.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$124.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$576.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$54.70
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$704.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb52.36
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$143.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$116.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$204.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb97.10
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb319.80

Tiffany Yeh

AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,005.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$525.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	\$12.54
Hon Precision (7769.TW)	O (04/17/2026)	NT\$6,425.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,805.00
Silicon Motion (SIMO.O)	O (05/06/2024)	\$253.60
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$6,250.00

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INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	\$476.15
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	\$17.54
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	\$41.50
Ambarella Inc (AMBA.O)	O (03/29/2016)	\$86.00
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	\$49.87
Analog Devices Inc. (ADI.O)	O (11/16/2023)	\$367.41
Astera Labs Inc (ALAB.O)	O (05/11/2025)	\$311.23
Broadcom Inc. (AVGO.O)	O (06/09/2024)	\$389.28
Cerebras Systems (CBRS.O)	O (06/08/2026)	\$198.71
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	\$49.99
Intel Corporation (INTC.O)	E (02/22/2023)	\$90.20
IonQ Inc (IONQ.N)	E (04/25/2023)	\$36.44
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	\$187.56
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	\$74.29
Micron Technology Inc. (MU.O)	O (10/06/2025)	\$823.03
Navitas Semiconductor Corp (NVT.S.O)	U (04/06/2025)	\$10.86
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	\$200.75
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	\$229.16
ON Semiconductor Corp. (ON.O)	++	\$81.61
Qorvo Inc (QRVO.O)	E (10/28/2025)	\$90.57
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	\$147.61
Quantinuum (QNT.O)	E (06/29/2026)	\$51.41

SanDisk Corporation. (SNDK.O)	O (03/03/2025)	\$1,214.83
Semtech Corp. (SMTC.O)	E (04/06/2025)	\$117.82
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	\$217.50
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	\$62.28
Texas Instruments (TXN.O)	U (04/13/2020)	\$275.74
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	\$23.62
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	\$239.69
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	\$340.02
Synopsys Inc. (SNPS.O)	E (02/27/2026)	\$388.76

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INDUSTRY COVERAGE: Semiconductor Capital Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Shane Brett		
Advanced Energy Industries Inc. (AEIS.O)	O (07/22/2026)	\$289.52
Applied Materials Inc. (AMAT.O)	E (05/18/2026)	\$507.67
Camtek (CAMT.O)	E (12/01/2025)	\$132.46
KLA Corp (KLAC.O)	O (01/15/2026)	\$182.82
Lam Research Corp (LRCX.O)	O (05/18/2026)	\$293.02
MKS Inc. (MKS.I.O)	O (08/04/2024)	\$297.45
Nova Ltd (NVMI.O)	E (12/01/2025)	\$390.87
ONTO Innovation Inc (ONTO.N)	O (06/14/2026)	\$258.58
Teradyne Inc (TER.O)	E (07/30/2025)	\$367.69

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